Analisis Statistik Deskriptif teerhadap IPK, Jam Belajar, dan Karakteristik Mahasiswa ITERA

FITRA

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1. Persiapan Lingkungan (Environment Setup)

Pada tahap ini, saya akan memuat semua library yang dibutuhkan dan mendefinisikan fungsi-fungsi statistik agar lebih mudah

```r
# ================================MEMUAT LIBRARY================================

library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(readr)

# ================================DEFINISI FUNGSI================================

# 1. Menghitung Skewness
skewness <- function(x) {
  x <- x[!is.na(x)]
  m <- mean(x)
  s <- sd(x)
  mean(((x - m) / s)^3)
}

# 2. Interpretasi
interpret_skewness <- function(value) {
  if (is.na(value)) return("Data tidak cukup untuk menghitung skewness.")

  if (value > 1) {
```

```r
    return("Distribusi sangat condong ke kanan (right-skewed kuat).")
  } else if (value > 0.5) {
    return("Distribusi condong ke kanan dengan tingkat moderat.")
  } else if (value > 0) {
    return("Distribusi sedikit condong ke kanan.")
  } else if (value == 0) {
    return("Distribusi simetris.")
  } else if (value > -0.5) {
    return("Distribusi sedikit condong ke kiri.")
  } else if (value > -1) {
    return("Distribusi condong ke kiri dengan tingkat moderat.")
  } else {
    return("Distribusi sangat condong ke kiri (left-skewed kuat).")
  }
}
```

2. Data Loading & Preprocessing

membaca dataset dan melakukan pembersihan data (Data Cleansing) sekaligus untuk semua kolom yang bermasalah. Ini memastikan data siap digunakan untuk visualisasi apa pun.

```r
# Load Data
data <- read.csv("Dataset/dataset.csv")

# --- DATA CLEANSING ---

# A. Pembersihan Kolom IPK Terakhir
# Mengganti karakter non-numerik dengan NA, memperbaiki desimal, dan filter range logis (0-4)
data$IPK.Terakhir..[data$IPK.Terakhir.. %in% c("-", ".", ",", "", " ")] <- NA
data$IPK.Terakhir.. <- gsub(",", ".", data$IPK.Terakhir..)
data$IPK.Terakhir.. <- as.numeric(data$IPK.Terakhir..)
```

```
## Warning: NAs introduced by coercion
```

```r
data$IPK.Terakhir..[data$IPK.Terakhir.. > 4 | data$IPK.Terakhir.. == 0] <- NA

# B. Pembersihan Kolom Jam Belajar
# Rename kolom, perbaikan desimal, dan filter range logis (0-60 jam/minggu)
names(data)[names(data) == "Rata.rata.belajar.perminggu..dalam.jam..."] <- "BelajarJam"
data$BelajarJam <- gsub(",", ".", data$BelajarJam)
data$BelajarJam <- as.numeric(data$BelajarJam)
```

```
## Warning: NAs introduced by coercion
```

```r
data$BelajarJam[data$BelajarJam < 0 | data$BelajarJam > 60] <- NA

# C. Pembersihan Kolom Anggota Keluarga
# Mengambil hanya angka, mengubah ke integer, batas wajar max 12
data$Jumlah.Anggota.Keluarga.. <- as.character(data$Jumlah.Anggota.Keluarga..)
data$Jumlah.Anggota.Keluarga.. <- gsub("[^0-9]", "", data$Jumlah.Anggota.Keluarga..)
```

```r
data$Jumlah.Anggota.Keluarga..[data$Jumlah.Anggota.Keluarga.. == ""] <- NA
data$Jumlah.Anggota.Keluarga.. <- as.integer(data$Jumlah.Anggota.Keluarga..)
data$Jumlah.Anggota.Keluarga..[data$Jumlah.Anggota.Keluarga.. > 12] <- NA

# D. Pembersihan Kolom Berat Badan
# Hapus "kg", spasi, ambil angka
data$Berat.Badan.. <- as.character(data$Berat.Badan..)
data$Berat.Badan.. <- gsub("kg", "", data$Berat.Badan.., ignore.case = TRUE)
data$Berat.Badan.. <- gsub("[^0-9.]", "", data$Berat.Badan..)
data$Berat.Badan.. <- as.numeric(data$Berat.Badan..)

# E. Pembersihan Kolom Tinggi Badan
# Hapus "cm", filter range logis (120-220 cm)
data$Tinggi.Badan.. <- as.character(data$Tinggi.Badan..)
data$Tinggi.Badan.. <- gsub("cm", "", data$Tinggi.Badan.., ignore.case = TRUE)
data$Tinggi.Badan.. <- gsub("[^0-9.]", "", data$Tinggi.Badan..)
data$Tinggi.Badan.. <- as.numeric(data$Tinggi.Badan..)
data$Tinggi.Badan..[data$Tinggi.Badan.. < 120 | data$Tinggi.Badan.. > 220] <- NA

# Menampilkan Ringkasan Data Bersih
summary(data)
```

```
##       NIM        Program.Studi..    IPK.Terakhir..  Jenis.Kelamin
##  Min.   :  1.0   Length:458         Min.   :1.400   Length:458
##  1st Qu.:115.2   Class :character   1st Qu.:3.100   Class :character
##  Median :229.5   Mode  :character   Median :3.400   Mode  :character
##  Mean   :229.5                      Mean   :3.373
##  3rd Qu.:343.8                      3rd Qu.:3.655
##  Max.   :458.0                      Max.   :4.000
##                                     NA's   :135
##  Tinggi.Badan..  Berat.Badan..    Pendidikan.terakhir   BelajarJam
##  Min.   :142.0   Min.   : 33.00   Length:458          Min.   : 0.00
##  1st Qu.:155.0   1st Qu.: 47.00   Class :character    1st Qu.: 3.00
##  Median :161.0   Median : 53.00   Mode  :character    Median : 7.00
##  Mean   :162.3   Mean   : 56.35                       Mean   :11.58
##  3rd Qu.:169.5   3rd Qu.: 62.00                       3rd Qu.:15.00
##  Max.   :192.0   Max.   :160.00                       Max.   :60.00
##  NA's   :7       NA's   :2                            NA's   :229
##  Apakah.penerima.beasiswa..
##  Length:458
##  Class :character
##  Mode  :character
##
##
##
##
##  Asal.Daerah..Kota.beserta.Provinsi.....Ex...Padang..Sumatera.Barat.
##  Length:458
##  Class :character
##  Mode  :character
##
##
##
```

```
##
##  Pekerjaan.saat.ini..selain.kuliah.
##  Length:458
##  Class :character
##  Mode  :character
##
##
##
##
##  Akses.Internet..yang.paling.sering.digunakan. Keterlibatan.Organisai
##  Length:458                                    Length:458
##  Class :character                              Class :character
##  Mode  :character                              Mode  :character
##
##
##
##
##  Uang.Saku.....yang.diberikan.oleh.orangtua.
##  Length:458
##  Class :character
##  Mode  :character
##
##
##
##
##  Jenis.Tempat.Tinggal.....Tempat.tinggal.saat.ini..
##  Length:458
##  Class :character
##  Mode  :character
##
##
##
##
##  Jarak.rumah.dari.kampus.ITERA.. Jenis.pekerjaan.Ayah.. Jenis.pekerjaan.Ibu..
##  Length:458                      Length:458             Length:458
##  Class :character                Class :character       Class :character
##  Mode  :character                Mode  :character       Mode  :character
##
##
##
##
##  Pendapatan.Orangtua.. Jumlah.Anggota.Keluarga..
##  Length:458            Min.   : 2.000
##  Class :character      1st Qu.: 4.000
##  Mode  :character      Median : 5.000
##                        Mean   : 4.665
##                        3rd Qu.: 5.000
##                        Max.   :10.000
##                        NA's   :4
```

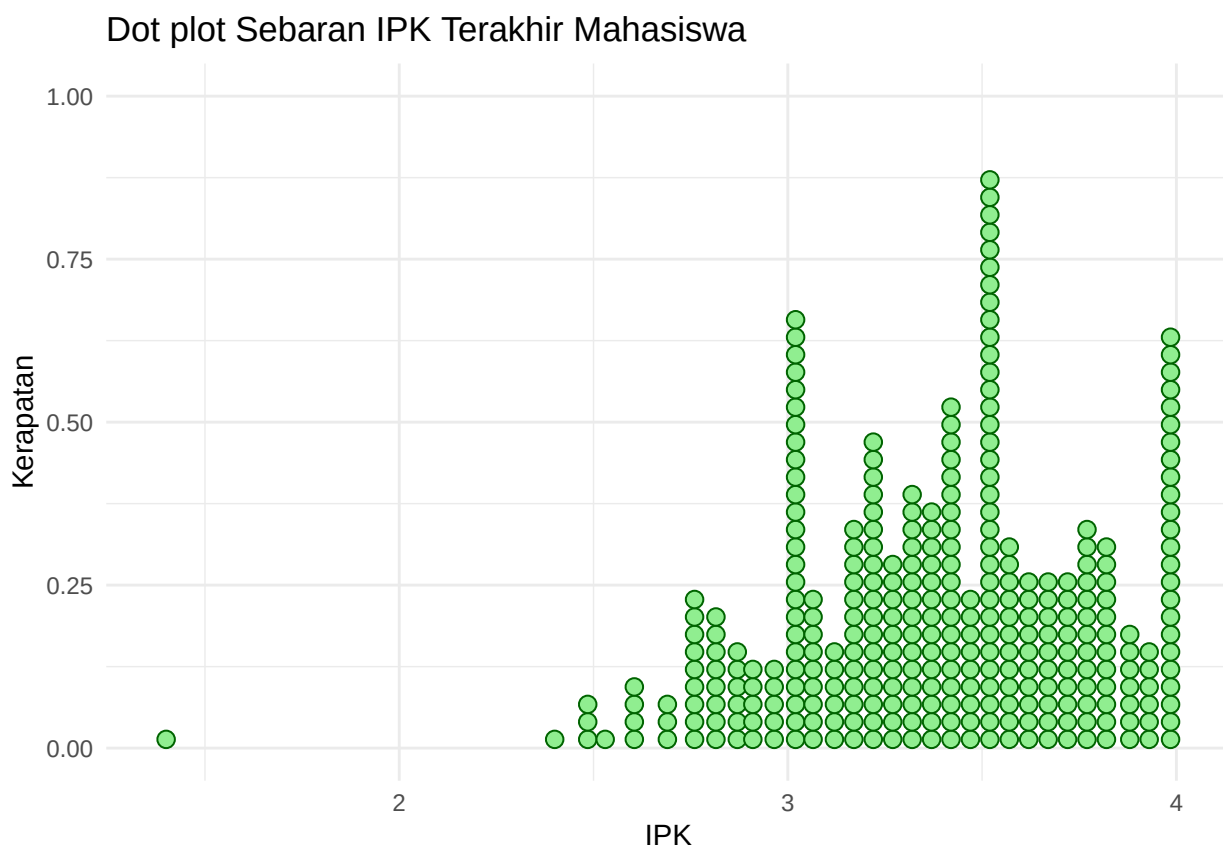
3. Analisis Variabel Akademik

```
A. Sebaran IPK Terakhir
```

visualisasi sebaran IPK mahasiswa menggunakan Dot Plot, stem-leaf, dan analisis Outlier menggunakan metode IQR.

```r
# 1. Dot Plot
ggplot(data, aes(x = `IPK.Terakhir..`)) +
  geom_dotplot(binwidth = 0.045, fill = "lightgreen", color = "darkgreen") +
  labs(
    title = "Dot plot Sebaran IPK Terakhir Mahasiswa",
    x = "IPK",
    y = "Kerapatan"
  ) +
  theme_minimal()
```

```
## Warning: Removed 135 rows containing missing values or values outside the scale range
## (`stat_bindot()`).
```




```r
# 2. Stem and Leaf Plot
cat("Stem and Leaf Plot IPK:\n")
```

```
## Stem and Leaf Plot IPK:
```

```r
stem(data$`IPK.Terakhir..`, scale = 2)
```

```
##
```

```
##   The decimal point is 1 digit(s) to the left of the |
##
##   14 | 0
##   15 |
##   16 |
##   17 |
##   18 |
##   19 |
##   20 |
##   21 |
##   22 |
##   23 |
##   24 | 07
##   25 | 003
##   26 | 00017
##   27 | 01455667888
##   28 | 00000033578999
##   29 | 0001257788
##   30 | 000000000000000111223333456667778
##   31 | 0034445567777889999
##   32 | 00000000011122244455666888999
##   33 | 000001112333334555556788889999
##   34 | 00000000000012233334577788999
##   35 | 0000000000000000000000000011334445556778888889
##   36 | 00000024445677788889
##   37 | 0000024444555668888999 9
##   38 | 000011133334667999
##   39 | 014445577
##   40 | 00000000000000000000
```

```
# 3. Deteksi Outlier (Metode IQR)
cat("==========================================================================")
```

```
## ==========================================================================
```

```
cat("\n")
```

```
Q1 <- quantile(data$`IPK.Terakhir..`, 0.25, na.rm = TRUE)
Q3 <- quantile(data$`IPK.Terakhir..`, 0.75, na.rm = TRUE)
IQR_value <- Q3 - Q1


batas_bawah <- Q1 - (1.5 * IQR_value)
batas_atas  <- Q3 + (1.5 * IQR_value)

cat("\n--- HASIL PERHITUNGAN IQR ---\n")
```

```
##
## --- HASIL PERHITUNGAN IQR ---
```

```
cat("Kuartil 1 (25%) :", Q1, "\n")
```

```
## Kuartil 1 (25%) : 3.1
```

```r
cat("Kuartil 3 (75%) :", Q3, "\n")
```

```
## Kuartil 3 (75%) : 3.655
```

```r
cat("Nilai IQR       :", IQR_value, "\n")
```

```
## Nilai IQR       : 0.555
```

```r
cat("Batas Bawah     :", batas_bawah, "\n")
```

```
## Batas Bawah     : 2.2675
```

```r
cat("Batas Atas      :", batas_atas, "\n")
```

```
## Batas Atas      : 4.4875
```

```r
cat("-----------------------------\n")
```

```
## -----------------------------
```

```r
data_outlier <- data[which(data$`IPK.Terakhir..` < batas_bawah |
                           data$`IPK.Terakhir..` > batas_atas), ]

if(nrow(data_outlier) > 0) {
  cat("DITEMUKAN", nrow(data_outlier), "DATA OUTLIER:\n")
  print(data_outlier[, c("IPK.Terakhir..")])
} else {
  cat("Data bersih. Tidak ditemukan outlier dengan metode IQR.\n")

}
```

```
## DITEMUKAN 1 DATA OUTLIER:
## [1] 1.4
```

```r
cat("==========================================================================\n")
```

```
## ==========================================================================
```

```r
cat("Ringkasan\n")
```

```
## Ringkasan
```

```r
summary(data$`IPK.Terakhir..`)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##   1.400   3.100   3.400   3.373   3.655   4.000     135
```

B. Distribusi Jam Belajar

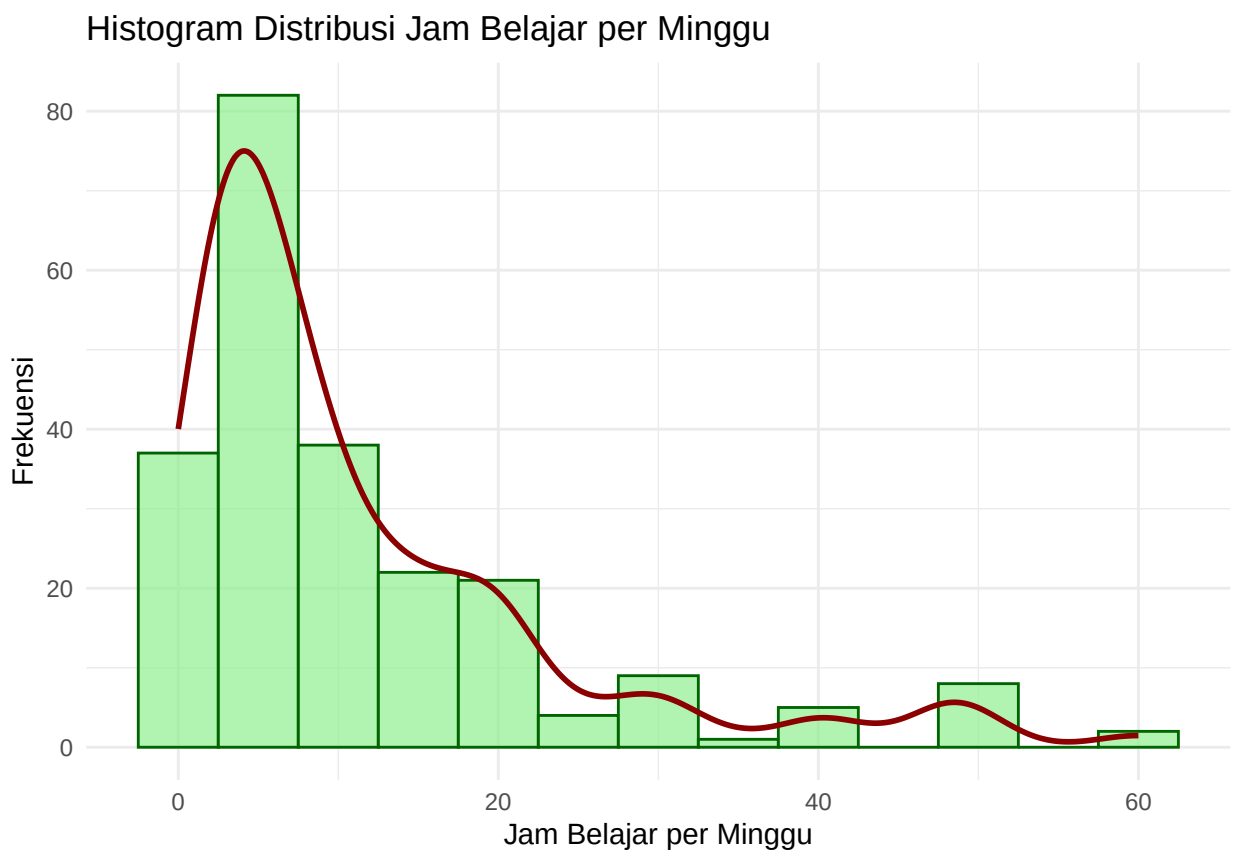
jam belajar mahasiswa per minggu.

```r
# 1. Histogram
names(data)[names(data) == "Rata.rata.belajar.perminggu..dalam.jam..."] <- "BelajarJam"
hist_plot <- ggplot(data, aes(x = BelajarJam)) +
  geom_histogram(
    binwidth = 5, # Anda bisa menyesuaikan lebar bin
    fill = "lightgreen",
    color = "darkgreen",
    alpha = 0.7
  ) +
  geom_density(aes(y = after_stat(count) * 5), color = "darkred", linewidth = 1) +
  labs(
    title = "Histogram Distribusi Jam Belajar per Minggu",
    x = "Jam Belajar per Minggu",
    y = "Frekuensi"
  ) +
  theme_minimal()

print(hist_plot)
```

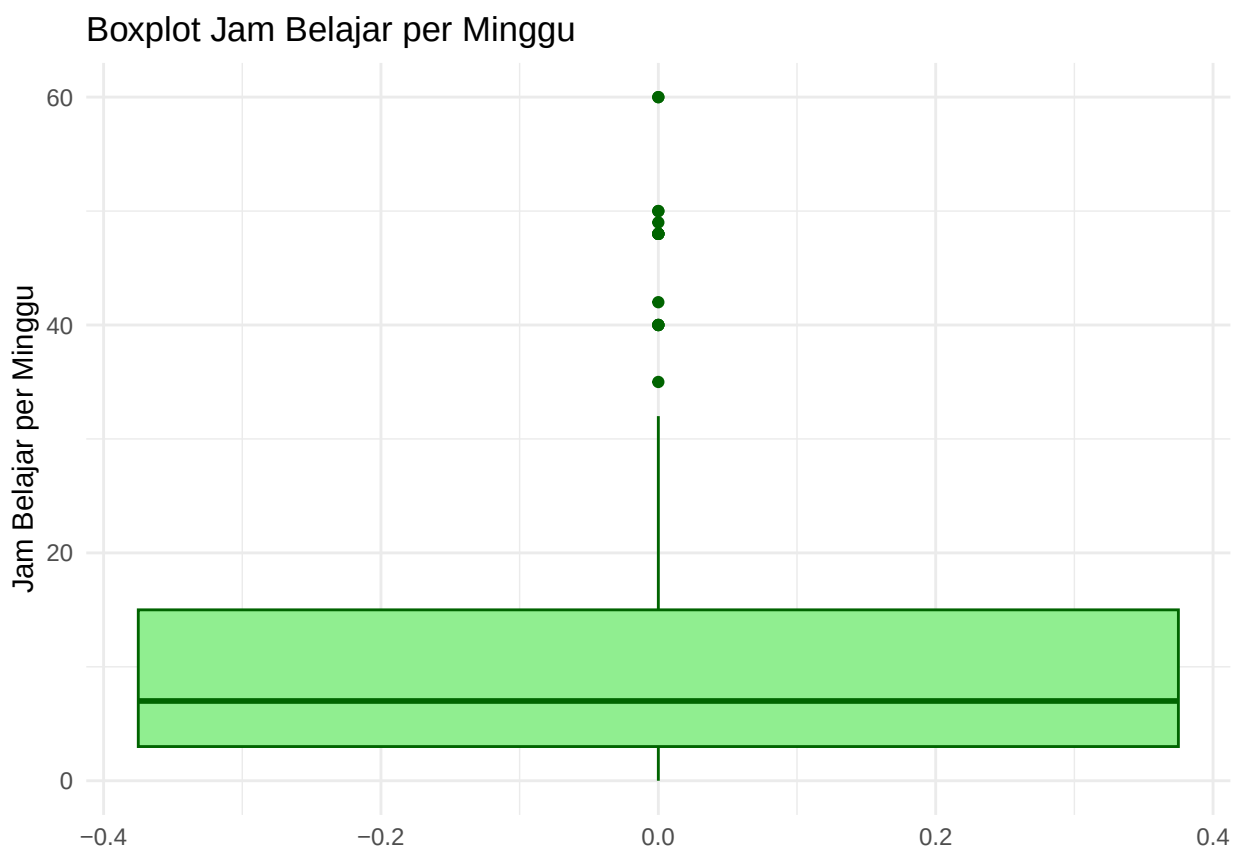
```
## Warning: Removed 229 rows containing non-finite outside the scale range
## (`stat_bin()`).

## Warning: Removed 229 rows containing non-finite outside the scale range
## (`stat_density()`).
```



```r
# 2. Boxplot
ggplot(data, aes(y = BelajarJam)) +
  geom_boxplot(fill = "lightgreen", color = "darkgreen") +
  labs(
    title = "Boxplot Jam Belajar per Minggu",
    y = "Jam Belajar per Minggu"
  ) +
  theme_minimal()
```

```
## Warning: Removed 229 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




```r
cat("Ringkasan\n")
```

```
## Ringkasan
```

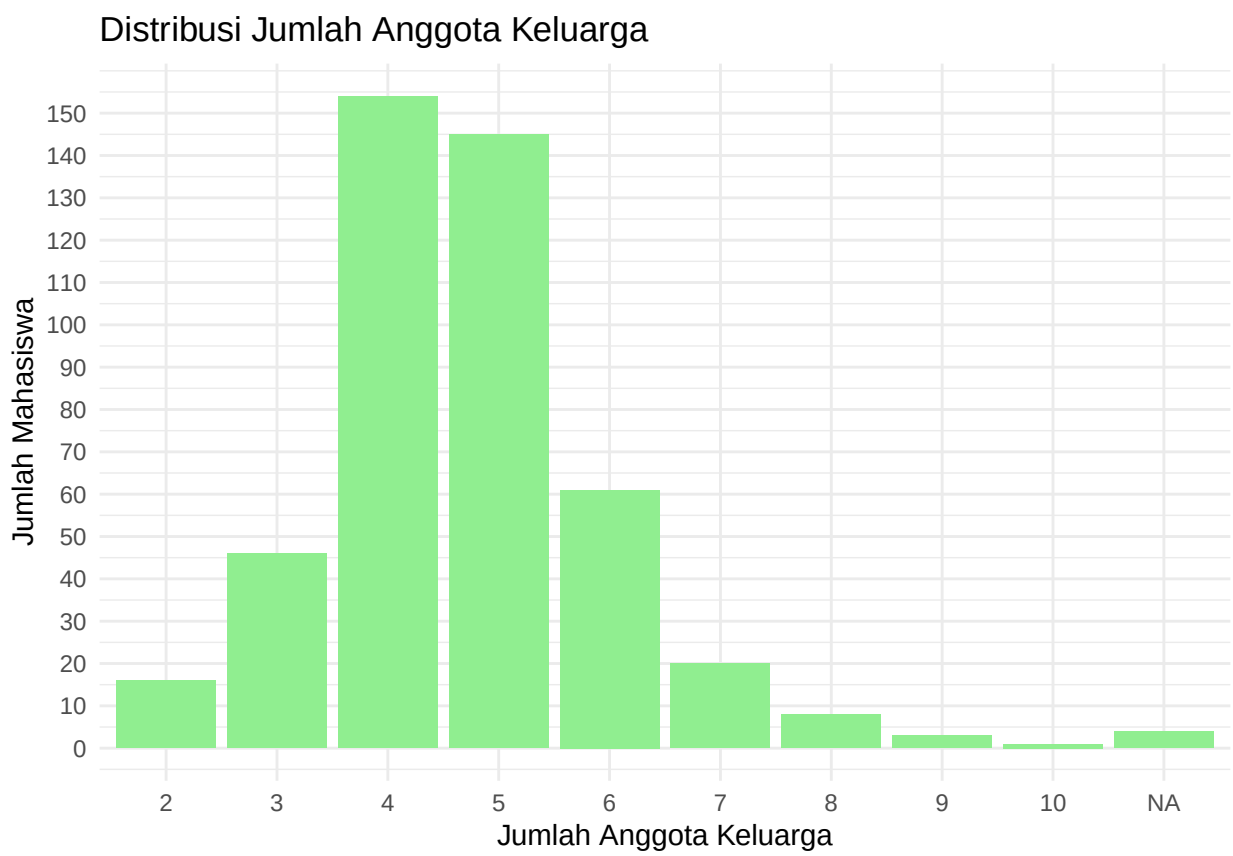
```r
summary(data$BelajarJam)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##    0.00    3.00    7.00   11.58   15.00   60.00     229
```

4. Analisis Variabel Demografis & Fisik

A. Jumlah Anggota Keluarga

```
ggplot(data, aes(x = factor(Jumlah.Anggota.Keluarga..))) +
  geom_bar(fill = "lightgreen") +
  scale_y_continuous(
    breaks = seq(0, max(table(data$Jumlah.Anggota.Keluarga..)), by = 10)
  ) +
  labs(
    title = "Distribusi Jumlah Anggota Keluarga",
    x = "Jumlah Anggota Keluarga",
    y = "Jumlah Mahasiswa"
  ) +
  theme_minimal()
```




```
summary(data$`Jumlah.Anggota.Keluarga..`)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##   2.000   4.000   5.000   4.665   5.000  10.000       4
```

B. Analisis Fisik (Berat & Tinggi Badan)

melihat kecendrungan (skewness) dari data fisik mahasiswa.

```r
# --- BERAT BADAN ---

cat("Ringkasan berat badan:\n")
```

```
## Ringkasan berat badan:
```

```r
summary(data$Berat.Badan..)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##   33.00   47.00   53.00   56.35   62.00  160.00       2
```

```r
cat("\n\nRingkasan tinggi badan:\n")
```

```
##
##
## Ringkasan tinggi badan:
```

```r
summary(data$Tinggi.Badan..)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##   142.0   155.0   161.0   162.3   169.5   192.0       7
```

```r
skew_berat <- skewness(data$Berat.Badan..)
cat("\n==========================================================================\n")
```

```
##
## ==========================================================================
```

```r
cat("--- ANALISIS BERAT BADAN ---\n")
```

```
## --- ANALISIS BERAT BADAN ---
```

```r
cat("Nilai Skewness :", round(skew_berat, 4), "\n")
```

```
## Nilai Skewness : 1.8729
```

```r
cat("Kesimpulan     :", interpret_skewness(skew_berat), "\n\n")
```

```
## Kesimpulan     : Distribusi sangat condong ke kanan (right-skewed kuat).
```

```r
# --- TINGGI BADAN ---
skew_tinggi <- skewness(data$Tinggi.Badan..)

cat("--- ANALISIS TINGGI BADAN ---\n")
```

```
## --- ANALISIS TINGGI BADAN ---
```

```r
cat("Nilai Skewness :", round(skew_tinggi, 4), "\n")
```

```
## Nilai Skewness : 0.267
```

```r
cat("Kesimpulan     :", interpret_skewness(skew_tinggi), "\n")
```

```
## Kesimpulan     : Distribusi sedikit condong ke kanan.
```